ENSy: An Interactive Instrument for Measuring How Dimensionality Reduction Distorts the Input-Embedding Geometry of Language Models

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Abstract

Dimensionality reduction is the usual way to inspect the embedding spaces of language models, but every projection to two or three dimensions distorts the neighbourhood structure it is meant to reveal. This work presents **ENSy (Embedding Navigation System)**, a browser-based instrument that projects the static input-embedding matrices of Qwen 2.5 0.5B-Instruct, Gemma 2 2B, GPT-2 and BERT base uncased with PCA, t-SNE, UMAP and Isomap, and quantifies each projection's distortion with the rank-based co-ranking framework: trustworthiness, continuity, $Q_{NX}(K)$, the behaviour indicator $B_{NX}(K)$ and a per-token decomposition $B_{NX,i}(K)$ shown in an interactive 3D viewer. The sample is a frozen, hashed set of $N = 438$ English word types from the Penn Treebank, balanced across nouns, verbs and function words and restricted to single-token forms in all four tokenizers. The implementation is validated against an independent library and synthetic rank permutations, and all results are regenerated by a pinned pipeline. The results indicate three patterns. First, the sign of t-SNE's behaviour indicator follows the perplexity: mildly intrusive or neutral at perplexity 5, extrusive at 30 in all four models, by margins well beyond the seed-to-seed variation. Second, every projection inflates the separation of the linguistic categories relative to the original space. Third, faithfulness differs between models independently of dimension: GPT-2 has the most faithful t-SNE and UMAP projections and BERT, with the same 768 dimensions, the least. The findings are descriptive and do not attribute causes to model architecture.

Keywords: word embeddings, dimensionality reduction, co-ranking matrix, trustworthiness, continuity, B_{NX}, t-SNE perplexity, UMAP, Isomap, PCA, visual analytics, reproducibility.

1. Introduction

The input-embedding matrix of a language model maps each token of its vocabulary to a vector in $\mathbb{R}^d$, with d in the hundreds or thousands, continuing the line of learned dense word representations [2]. Because such spaces cannot be inspected directly, analyses of their geometry routinely rely on two- or three-dimensional projections, whose characteristic distortions are well documented in visual analytics [3]. A projection, however, cannot preserve every neighbourhood relation of the original space; points that are far apart may be drawn together (*intrusions*) and genuine neighbours may be pulled apart (*extrusions*). Without measuring these distortions, a reader cannot tell which visual clusters belong to the model and which were created by the projection.

ENSy addresses this gap. It is not the object of study of the thesis but the instrument that produces its evidence: a reproducible pipeline that projects embedding matrices with four standard algorithms and reports, for every projection, how faithfully it preserves K-ary neighbourhoods, together with an interactive 3D viewer

in which those distortions can be inspected token by token.

The guiding question is: *how do PCA, t-SNE, UMAP and Isomap preserve or distort the neighbourhood structure of the input-embedding spaces of language models?* The contributions are:

1. **A validated co-ranking implementation** — trustworthiness/continuity identical to an independent implementation, the behaviour indicator $B_{NX}(K)$ checked against the primary source and anchored empirically, and an exact per-token decomposition $B_{NX,i}(K)$ (Section 3.6).
2. **A frozen experimental protocol** — a hashed Penn-Treebank-derived lexical dataset, pinned model revisions, a fully explicit parameter table and a continuous-integration run that rebuilds the dataset and reproduces the results (Section 3).
3. **An interactive instrument** — a WebGL viewer with a reference-partition colouring, a thermal camera for $B_{NX,i}(K)$ and neighbourhood probing (Section 4).
4. **Empirical results** for 4 models × 5 projection configurations, a seed/perplexity sensitivity analysis and a PCA centring ablation (Section 5).

2. Background

2.1 Input embeddings and cosine geometry

ENSy studies exclusively the static input-embedding matrix of each model: the lookup table applied before any contextual layer. This representation is directly comparable across decoder-only and encoder-only architectures, whereas contextual hidden states would require choices of layer, context and pooling that change what is being measured. Similarity between two embeddings $u, v \in \mathbb{R}^d$ is measured by the cosine of the angle θ between them,

$$\cos\theta = \frac{u^\top v}{\|u\|\,\|v\|},$$

and cosine distance is $1 - \cos\theta$. All vectors are L_2-normalised, $\hat{u} = u/\|u\|$, so that

$$\|\hat{u} - \hat{v}\|^2 = 2\,(1 - \cos\theta),$$

that is, Euclidean distance on the unit sphere is a monotone function of cosine distance. Normalisation does not remove the anisotropy of embedding spaces [4], [5]; it only makes cosine and Euclidean neighbourhoods coincide. As a consequence, the neighbour ranks used by every metric below are identical whichever of the two metrics is used in the original space.

2.2 Projection algorithms

PCA projects onto the directions of maximum variance of the centred data, computed by a singular value decomposition $X_c = U\Sigma V^\top$, $Y = X_c V_{1:3}$ [6], [7]. A non-centred variant (PCA-NC) is used only as an ablation.

t-SNE matches neighbour distributions with a Gaussian kernel in the original space and a Student-t kernel in the projection by minimising $\mathrm{KL}(P \,\|\, Q)$ [8]. Its *perplexity* is "a smooth measure of the effective number of neighbors", with "typical values between 5 and 50" [8, p. 2582].

UMAP builds a fuzzy neighbourhood graph and optimises a low-dimensional layout by cross-entropy [9]; its number of neighbours sets "the local scale at which to approximate the manifold".

Isomap estimates geodesic distances as shortest paths in a k-nearest-neighbour graph and embeds them with classical MDS [10]. Points disconnected from the graph's giant component have no finite geodesic distance; the original authors note they "can be deleted from further analysis" [10, note 19].

2.3 The co-ranking framework

Let ρ_{ij} be the rank of j among the neighbours of i in the original space and r_{ij} its rank in the projection (reflexive ranks are 0; ties are broken by index). The co-ranking matrix [1] is the joint histogram of the two ranks,

$$Q = [q_{kl}]_{1 \le k,l \le N-1},$$
$$q_{kl} = \big|\{(i,j) : \rho_{ij} = k \text{ and } r_{ij} = l\}\big|,$$

with rows indexed by the original rank. For a neighbourhood size K, Q splits into the blocks UL_K, UR_K, LL_K, LR_K, and UL_K into its diagonal D_K and lower/upper triangles LT_K, UT_K. *Mild* K-intrusions (items promoted inside the neighbourhood) fall in LT_K, mild K-extrusions (items demoted inside it) in UT_K, and *hard* intrusions and extrusions — pairs that cross the K boundary — in LL_K and UR_K.

Trustworthiness and continuity [11] penalise hard intrusions and hard extrusions, weighted by how far they cross the boundary:

$$M_T(K) = 1 - \frac{2}{G_K} \sum_{(k,l) \in LL_K} (k - K)\, q_{kl},$$

$$M_C(K) = 1 - \frac{2}{G_K} \sum_{(k,l) \in UR_K} (l - K)\, q_{kl},$$

with $G_K = NK(2N - 3K - 1)$ for $K < N/2$.

Overall quality and behaviour. Following Lee and Verleysen [1, Eqs. 22–25],

$$U_N(K) = \frac{1}{KN} \sum_{LT_K} q_{kl}, \qquad U_X(K) = \frac{1}{KN} \sum_{UT_K} q_{kl},$$

$$U_P(K) = \frac{1}{KN} \sum_{D_K} q_{kl}, \qquad Q_{NX}(K) = U_P + U_N + U_X,$$

$$B_{NX}(K) = U_X(K) - U_N(K).$$

$Q_{NX}(K)$ is the average fraction of shared K-neighbours and equals the local continuity meta-criterion of Chen and Buja [12] up to the baseline $K/(N-1)$. An earlier conference version of the framework [13] attaches the names U_N and U_X to the opposite triangles while yielding the same value of B_{NX}; the definitions above follow the journal version. $B_{NX}(K)$ is, in the authors' words, "positive for intrusive embeddings and negative for extrusive ones" [1, p. 1434]. The sign looks inverted only at first sight: a single hard intrusion pushes every subsequent genuine neighbour one rank down, producing $(K - r_{ij})$ mild extrusions, so the number of mild extrusions measures the severity of the intrusions [1, Fig. 2]. The same paper defines a second behaviour indicator, $B_{TC}(K) = M_C(K) - M_T(K)$ (Eq. 27), which weights hard events instead of counting mild ones; both are reported.

Per-token decomposition. Since $q_{kl} = \sum_i q_{kl}^{(i)}$, applying the same masks to the counts of a single point i gives

$$B_{NX,i}(K) = \frac{1}{K}\Big(\sum_{UT_K} q_{kl}^{(i)} - \sum_{LT_K} q_{kl}^{(i)}\Big),$$

$$B_{NX}(K) = \frac{1}{N} \sum_{i=1}^{N} B_{NX,i}(K),$$

with $|B_{NX,i}(K)| \le (K-1)/K$. ENSy's thermal camera displays this quantity, which is the token's net contribution to intrusive or extrusive behaviour, not a general "local distortion".

2.4 Silhouette

For the reference partition (Section 3.3), the Silhouette $s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}$ [14] measures how well the categories are separated. It is computed in the projected space, in both 3D and 2D, and — as a reference — in the original space with the cosine metric. It measures separation, not its cause.

3. Materials and Methods

3.1 Models

Four pre-trained checkpoints are used — Qwen 2.5 [15], Gemma 2 [16], GPT-2 [17] and BERT [18] — pinned to the Hugging Face revisions recorded in the dataset file; tokenizers are loaded with the Transformers library [19]:

Model	Repository	Type	d	Vocabulary
Qwen 2.5 0.5B-Instruct	`Qwen/Qwen2.5-0.5B-Instruct`	decoder	896	151,665
Gemma 2 2B	`google/gemma-2-2b`	decoder	2,304	256,000
GPT-2 (124M)	`openai-community/gpt2`	decoder	768	50,257
BERT base uncased	`bert-base-uncased`	encoder	768	30,522

GPT-2 [17] is included as a model trained differently from the other three: an earlier generation, trained on a different corpus (WebText) with a different recipe, and the model most studied in earlier work on embedding geometry [4]. The models differ simultaneously in architecture, dimension, vocabulary, objective, corpus and training recipe; the design allows observing whether distortion patterns hold across models, not attributing them to architecture.

3.2 Extraction protocol

For each word w, the token looked up is the word-initial surface form — `" " + w` for the byte-level BPE (Qwen, GPT-2) and SentencePiece (Gemma) tokenizers, w for WordPiece (BERT). The distinction matters: in Qwen, `run` and `run` are different rows (ids 6108 and 1598). A word is included only if, in every tokenizer, it encodes to exactly one id without special tokens, the id is not the unknown or a special token, the id is a valid row of the embedding matrix, and it decodes back to the word.

The selected rows are read directly from the checkpoint's safetensors file with HTTP range requests: only the file header and the required rows are transferred (kilobytes to a few megabytes instead of multi-gigabyte checkpoints). The rows are bit-identical to `model.get_input_embeddings().weight` and to a full download (tested for Qwen and BERT). They are cast to `float32` and L_2-normalised.

3.3 Lexical dataset derived from the Penn Treebank

The dataset is built from the freely distributed sample of the Penn Treebank [20] in NLTK [21] (the full treebank is licensed by the LDC). Tokens are lower-cased; non-alphabetic tokens and proper nouns (`NNP/NNPS`) are discarded. Tags map to a *reference linguistic partition*: `NN, NNS` → `content_noun`; `VB*` → `content_verb`, except the auxiliary forms of *be*, *have* and *do*, which count as `function_word`; `DT, PDT, IN, CC, TO, MD, PRP, PRP`, WDT, WP, WP`, WRB, EX, RP` → `function_word`. Two fixed rules remove tokenisation artefacts: clitic stems (`ca`, `wo`, `ai` from `ca n't`) and function-word forms internal to foreign names (`de`, `la`, `le`, `del`). Each form receives its majority category over all its occurrences; ties and forms whose majority lies outside the partition are excluded. After the single-token filter, each category is cut to the size of the smallest one, keeping its most frequent forms.

Stage	Count
PTB tokens (NLTK sample)	100,676
Lexical types (lower-cased, alphabetic, no proper nouns)	7,295
Excluded artefacts (clitic stems, name-internal forms)	7
Excluded: category tie / majority outside the partition	78 / 1,418
After disambiguation	5,792
Dropped as multi-token (first failing tokenizer)	Qwen 2.5 0.5B 932, GPT-2 61, Gemma 2 2B 4, BERT base 267
Single-token in all four tokenizers: noun / verb / function	2,672 / 1,710 / 146
Balanced dataset N	**438** (146 per category)

Dataset hash `c2416895b599`.

The function words are the binding category; the final N is reported as obtained, not adjusted to a target. The file `data/dataset.json` records the rules, the model revisions and a SHA-256 hash of the items.

3.4 Projection parameters

PCA, t-SNE and Isomap use scikit-learn [22] and UMAP its reference implementation [9]. All parameters were fixed before collection; library defaults are written out so that results do not depend on the installed version's defaults.

Algorithm	Parameters
PCA	3 components, centred, full SVD; component sign fixed (largest loading positive)
PCA-NC (ablation)	as PCA, without centring
t-SNE	perplexity 5 (frozen) **and** 30 (library default); cosine; seed 42; PCA initialisation; learning rate "auto"; 1000 iterations; Barnes–Hut, $\theta = 0.5$
UMAP	15 neighbours, min. distance 0.1 (library defaults); cosine; seed 42
Isomap	15 neighbours; cosine; giant component of the k-NN graph only

t-SNE is reported at both perplexities because the sensitivity analysis (Section 5.4) shows that the sign of its behaviour indicator depends on this choice. Each configuration is also run with two components to compute the 2D Silhouette.

3.5 Evaluation protocol

For each of the 4 × 5 configurations, M_T, M_C, Q_{NX}, B_{NX} and B_{TC} are computed for $K \in \{5, 10, 15, 20\}$ and presented as curves; $K = 10$ is the reference value, following the observation that "a human analyst usually looks at a handful (say 10) data points around" a point of interest [23, p. 894]. Ranks in the original space use the cosine distance; ranks in the projection use the Euclidean distance. Two analyses are kept separate from the main matrix: a **sensitivity analysis** (seeds 0, 1, 2, 3 and 42 for t-SNE at both perplexities and for UMAP) and a **PCA-NC ablation**.

Reproducibility. The canonical run is a GitHub Actions workflow on a clean checkout with pinned dependencies (CPU-only PyTorch). It runs the validation suite, rebuilds the dataset and checks that its hash matches the committed one, runs the main matrix, the ablation and the sensitivity analysis, and compares its numbers with the committed results.

3.6 Validation of the implementation

Before any result was produced, the metric code was validated by an automated suite:

- M_T and M_C coincide with `sklearn.manifold.trustworthiness` [22] (applied in both directions) to 10^{-12} for every K;
- every row and column of Q sums to N, and hard intrusions equal hard extrusions;
- **synthetic rank permutations**: identity gives $B_{NX} = 0$ and $M_T = M_C = Q_{NX} = 1$; promoting one item from rank m to rank 1 gives exactly $B_{NX,i} = (m-2)/K > 0$; demoting rank 1 to m gives $-(m-2)/K$; a single swap gives $U_N = U_X = 1/(KN)$;
- **sign anchor**: a sphere crushed by orthographic projection (a textbook intrusive map) yields $B_{TC} > 0$ and $B_{NX} > 0$; a sphere torn open yields both negative;
- the mean of $B_{NX,i}(K)$ equals $B_{NX}(K)$ to 10^{-12}, and $|B_{NX,i}| \leq (K-1)/K$.

4. The ENSy instrument

4.1 Architecture

A FastAPI server loads the frozen dataset, extracts and caches the normalised rows per model, and serves, for each (model, algorithm) pair, the projected coordinates, the metric curves, $B_{NX,i}(K)$ for every token and K, the 20 nearest neighbours of every token in both spaces, and both Silhouette scores. A React/TypeScript client renders the projection with Three.js. No placeholder data is ever displayed: when a projection is being computed the viewer says so, and when data are unavailable it shows the reason. The screenshots in this manuscript were captured from a local (Windows) server; the t-SNE and UMAP values visible in their panels can therefore differ slightly from the canonical values of Section 5 (see Section 6.1).

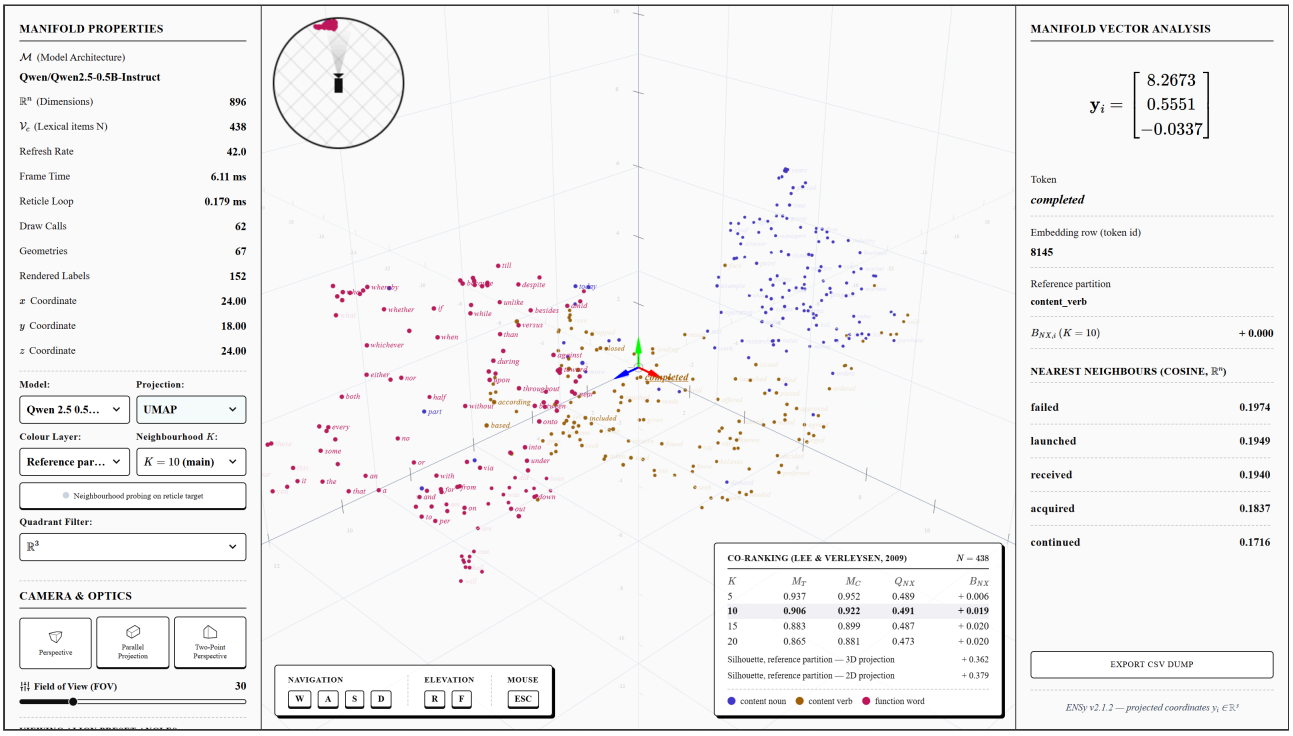


Fig. 1. The ENSy workspace (Qwen 2.5, UMAP, isometric view). Left: model, projection, colour layer and K selectors, camera optics and live profiling. Centre: the 438 tokens, coloured by the reference partition. Bottom right: the co-ranking curves of the current projection. Right: the inspector for the token under the reticle, with its true, false and missing neighbours.

4.2 Faithful rendering

The rendered cloud is the projection itself: coordinates are placed with a single isotropic similarity transform (median centre, one scale for all axes, set by the 99th-percentile radius), which preserves all ratios of distances. All tokens live in one `BufferGeometry` with per-vertex RGBA colour (one draw call for the cloud regardless of N). Every token is drawn as a dot. Words are placed as in the TensorFlow Embedding Projector: each frame, labels are written to the right of their dots, nearest tokens first, wherever they do not overlap a label already placed; a token whose word does not fit keeps only its dot, and flying closer spreads the points so that more words fit.

4.3 Colour layers and probing

Three layers share the colour channel with a fixed precedence — probing > thermal camera > reference partition — and never mix on one token. The **reference partition** colours nouns, verbs and function words. The **thermal camera** maps $B_{NX,i}(K)$ onto a fixed bipolar scale on $[-1, 1]$ (cyan: extrusive; light grey: neutral; orange: intrusive), never normalised per frame, so equal values always receive equal colours across projections. In this layer the colour is carried by the dot and the word stays dark, because most local values are close to zero and would otherwise give pale, illegible words. **Probing** colours, for the token under the reticle, its true ($N_K \cap n_K$, green), false ($n_K \setminus N_K$, red) and missing ($N_K \setminus n_K$, translucent blue) neighbours.

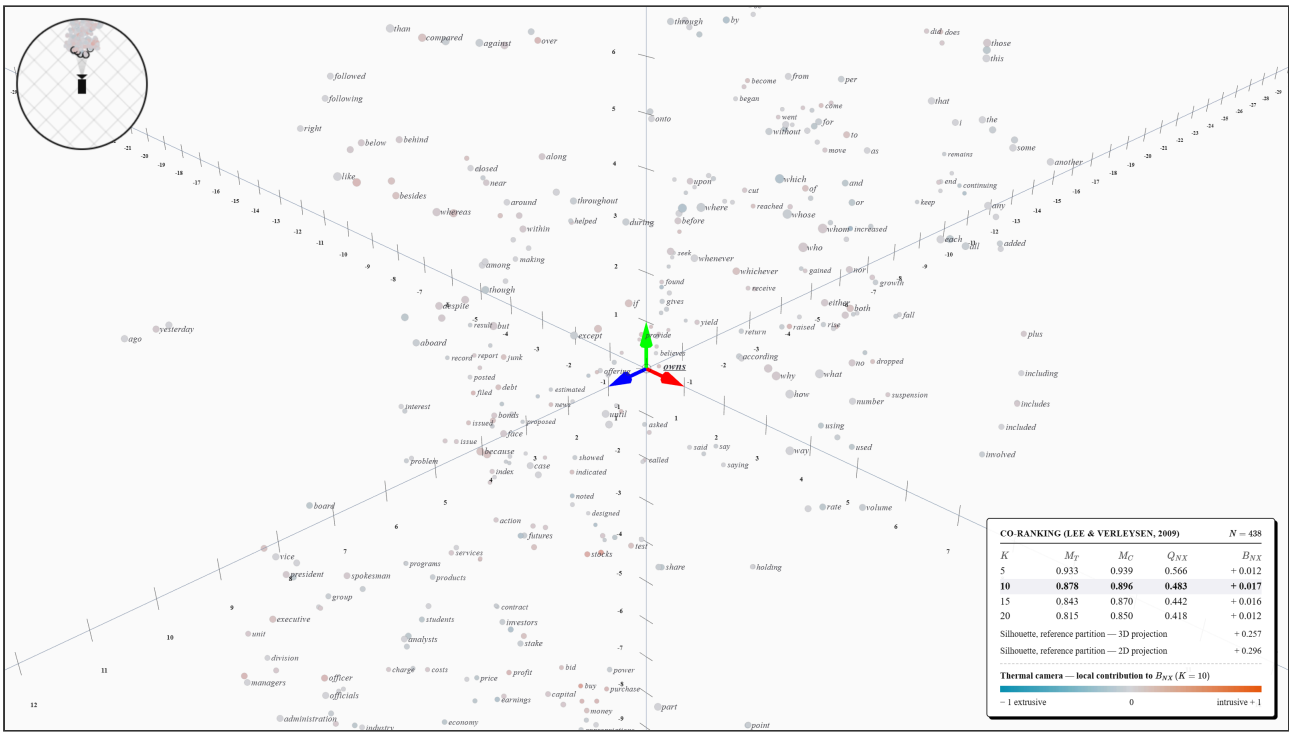


Fig. 2. Thermal camera on Qwen 2.5 with t-SNE (perplexity 5), $K = 10$ (viewport only, grid planes off): each token's local contribution to $B_{NX}(10)$ on the fixed scale $[-1, 1]$.

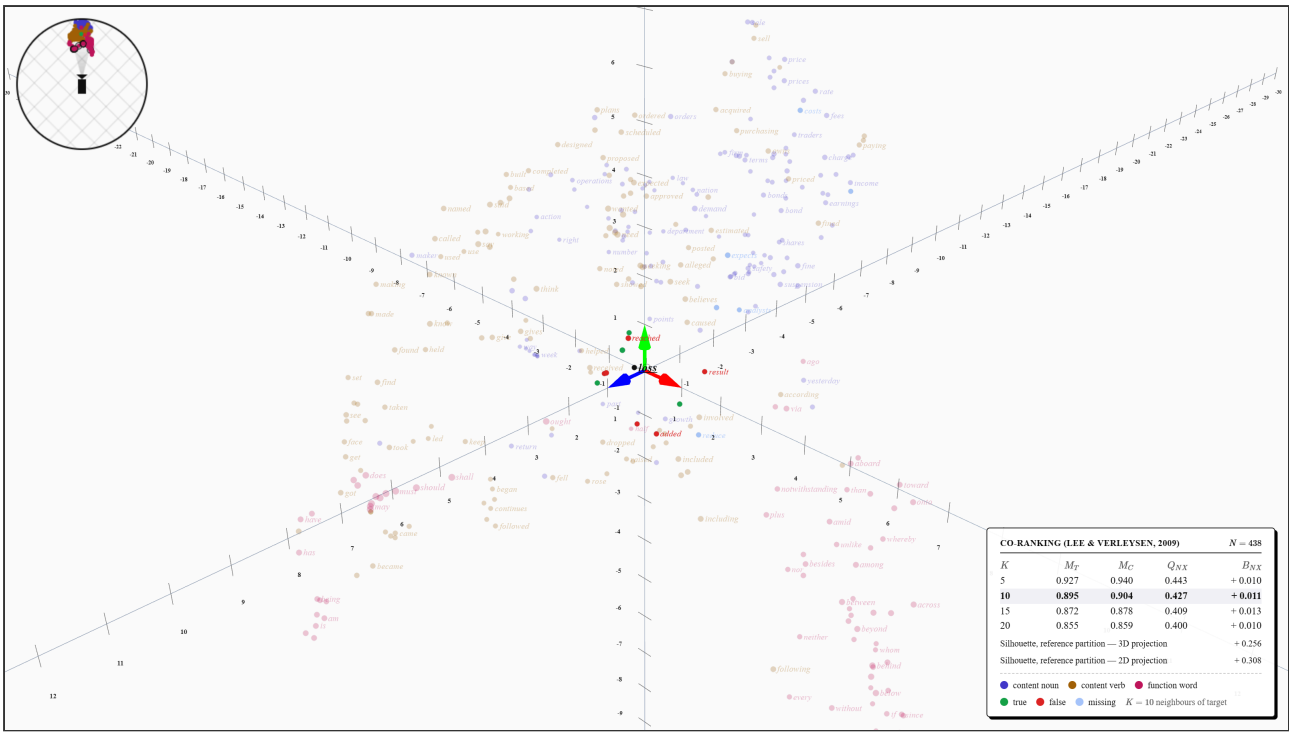


Fig. 3. Neighbourhood probing after flying towards a token (BERT, UMAP, $K = 10$; viewport only, grid planes off). Green: genuine neighbours kept by the projection; red: false neighbours introduced by it; translucent blue: genuine neighbours it pushed away.

4.4 Navigation and interface

The viewer offers first-person 6-DOF flight (W/A/S/D, R/F, mouse look under pointer lock), perspective, parallel and two-point perspective cameras with LERP/SLERP transitions between front, top, side and isometric presets, an octant filter and a minimap. A central reticle selects the nearest token in screen space within 25 px.

CO-RANKING (LEE & VERLEYSEN, 2009) $N = 438$

K	M_T	M_C	Q_{NX}	B_{NX}
5	0.831	0.929	0.184	− 0.014
10	**0.829**	**0.927**	**0.266**	**+ 0.008**
15	0.823	0.925	0.320	+ 0.021
20	0.825	0.915	0.352	+ 0.035
Silhouette, reference partition — 3D projection				+ 0.315
Silhouette, reference partition — 2D projection				+ 0.397

content noun content verb function word

Fig. 4. Co-ranking panel for GPT-2 with Isomap: M_T, M_C, Q_{NX} and B_{NX} over K, and the 3D and 2D Silhouette scores.

5. RESULTS

5.1 Dataset

The pipeline yields $N = 438$ word types (146 per category); the function words are the binding category. In the table, each multi-token rejection is attributed to the first tokenizer that rejected it, in the order Qwen, GPT-2, Gemma 2, BERT.

5.2 Neighbourhood preservation

Model	Algorithm	M_T	M_C	Q_{NX}	B_{NX}	B_{TC}	s_{3D}	s_{2D}
Qwen 2.5 0.5B	PCA	0.830	0.828	0.205	−0.007	−0.001	+0.161	+0.081
Qwen 2.5 0.5B	t-SNE ($p = 5$)	0.875	0.891	0.477	+0.016	+0.016	+0.243	+0.296
Qwen 2.5 0.5B	t-SNE ($p = 30$)	0.890	0.883	0.479	−0.041	−0.008	+0.153	+0.330
Qwen 2.5 0.5B	UMAP	0.899	0.923	0.486	+0.032	+0.025	+0.370	+0.387
Qwen 2.5 0.5B	Isomap	0.792	0.910	0.248	+0.028	+0.117	+0.281	+0.341
GPT-2	PCA	0.827	0.846	0.216	−0.015	+0.019	+0.236	+0.243
GPT-2	t-SNE ($p = 5$)	0.896	0.914	0.457	+0.027	+0.017	+0.254	+0.312
GPT-2	t-SNE ($p = 30$)	0.912	0.912	0.495	−0.024	+0.000	+0.205	+0.286
GPT-2	UMAP	0.917	0.938	0.474	+0.012	+0.021	+0.363	+0.402
GPT-2	Isomap	0.829	0.927	0.266	+0.008	+0.098	+0.315	+0.397
Gemma 2 2B	PCA	0.867	0.802	0.189	−0.010	−0.065	+0.240	+0.157
Gemma 2 2B	t-SNE ($p = 5$)	0.884	0.879	0.438	+0.013	−0.005	+0.234	+0.292
Gemma 2 2B	t-SNE ($p = 30$)	0.901	0.909	0.432	−0.027	+0.008	+0.238	+0.283
Gemma 2 2B	UMAP	0.911	0.934	0.428	+0.005	+0.022	+0.345	+0.363
Gemma 2 2B	Isomap	0.864	0.920	0.238	+0.012	+0.057	+0.298	+0.336
BERT base	PCA	0.819	0.810	0.182	−0.005	−0.008	+0.159	+0.089
BERT base	t-SNE ($p = 5$)	0.876	0.853	0.429	+0.001	−0.023	+0.201	+0.262
BERT base	t-SNE ($p = 30$)	0.874	0.847	0.405	−0.039	−0.027	+0.181	+0.222
BERT base	UMAP	0.895	0.903	0.425	+0.014	+0.008	+0.261	+0.321
BERT base	Isomap	0.794	0.860	0.201	+0.012	+0.066	+0.166	+0.056

Source: `results/main_c2416895b599.json` *— commit* `e7f5a91f` *(clean working tree), Python 3.12.14, scikit-learn 1.6.1, umap-learn 0.5.7, seed 42, extraction: HTTP Range rows.*

s_{3D}, s_{2D}: Silhouette of the reference partition in the 3D and 2D projections. For reference, the Silhouette in the original space and the variance captured by three principal components:

Model	d	s in $\mathbb{R}^d$ (cosine)	PCA variance, 3 components
Qwen 2.5 0.5B	896	+0.031	7.8%
GPT-2	768	+0.048	10.7%
Gemma 2 2B	2304	+0.044	10.9%
BERT base	768	+0.025	11.0%

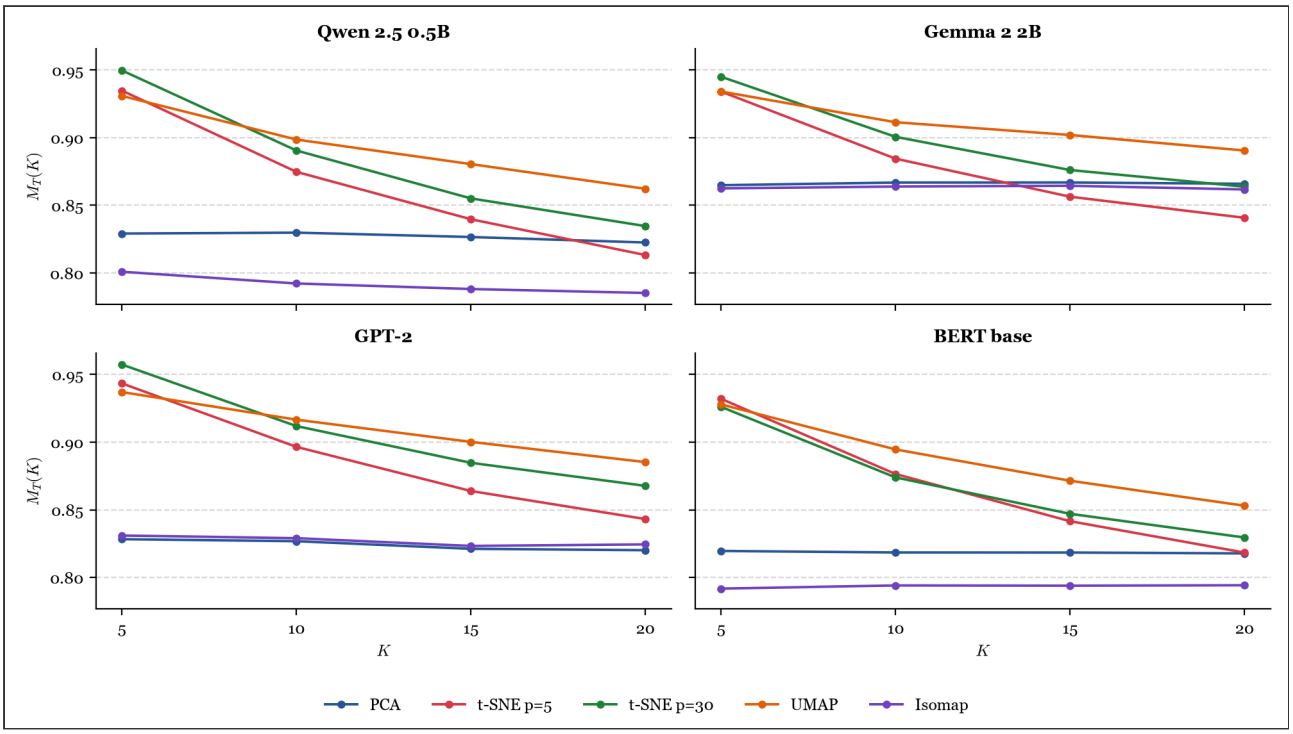


Fig. 5. Trustworthiness $M_T(K)$ over K for every model and projection configuration.

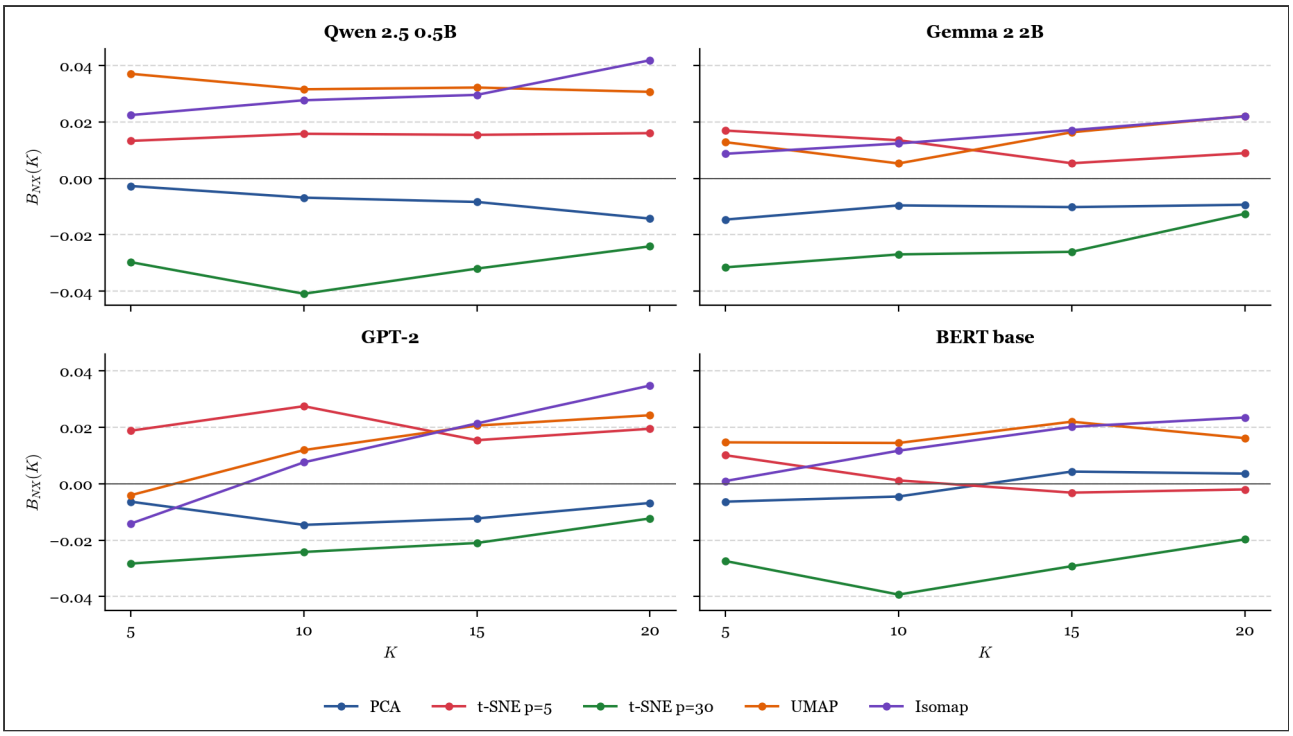


Fig. 6. Behaviour indicator $B_{NX}(K)$ over K (positive: intrusive; negative: extrusive). With perplexity 30, t-SNE is below zero at every K in all four models; with perplexity 5 it is above zero for Qwen, Gemma 2 and GPT-2 and close to zero for BERT.

5.3 Visual comparison

Figures 7–11 show the five projection configurations of the same model, and Figures 12–15 the same algorithm (UMAP) across the four models. All are seen from the same camera (isometric preset, then a one-second forward flight along the aim), with the grid planes switched off. Colours: indigo nouns, ochre verbs, magenta function words.

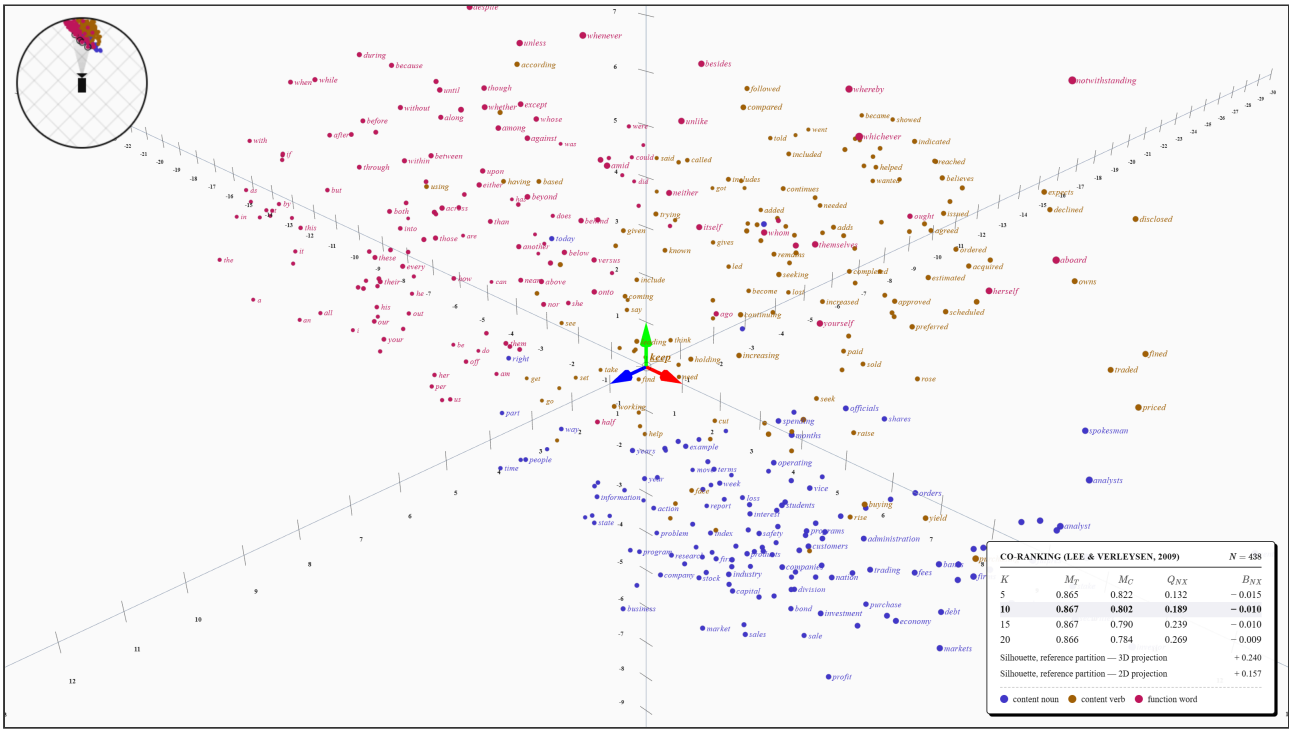


Fig. 7. Gemma 2 2B — PCA (centred).

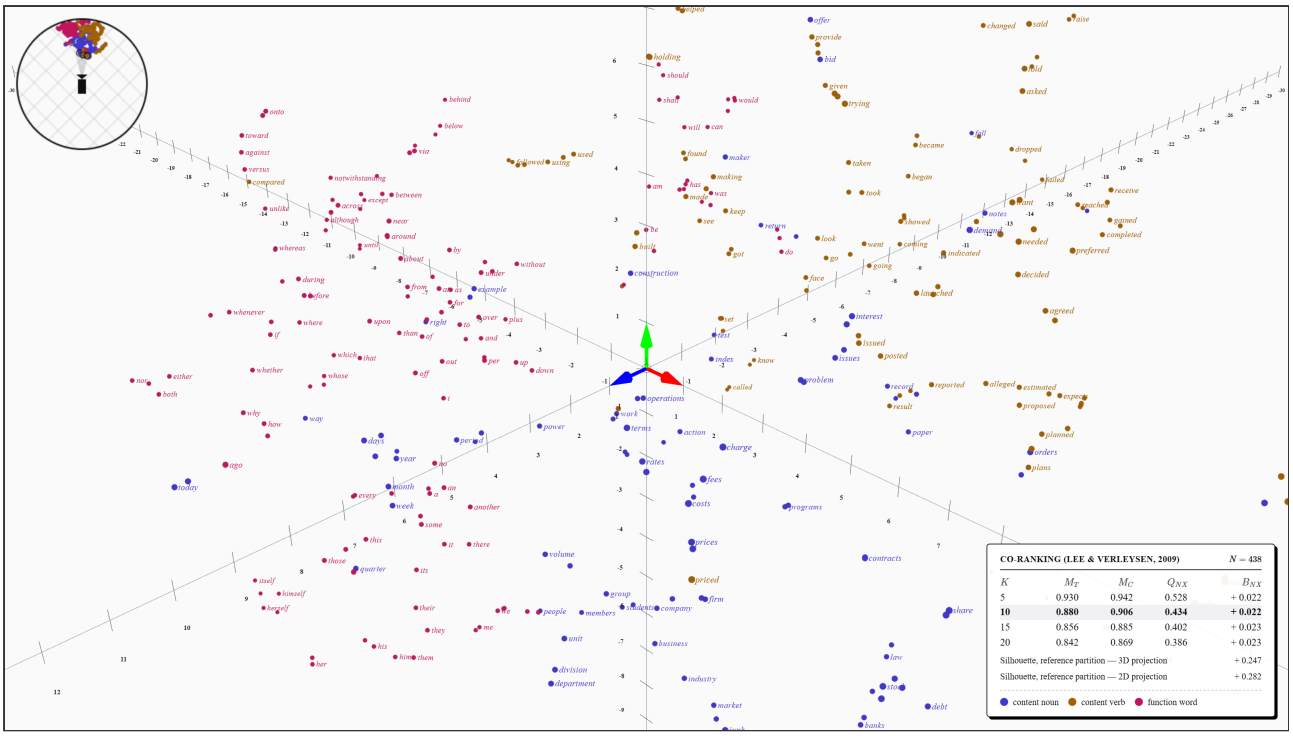


Fig. 8. Gemma 2 2B — t-SNE, perplexity 5.

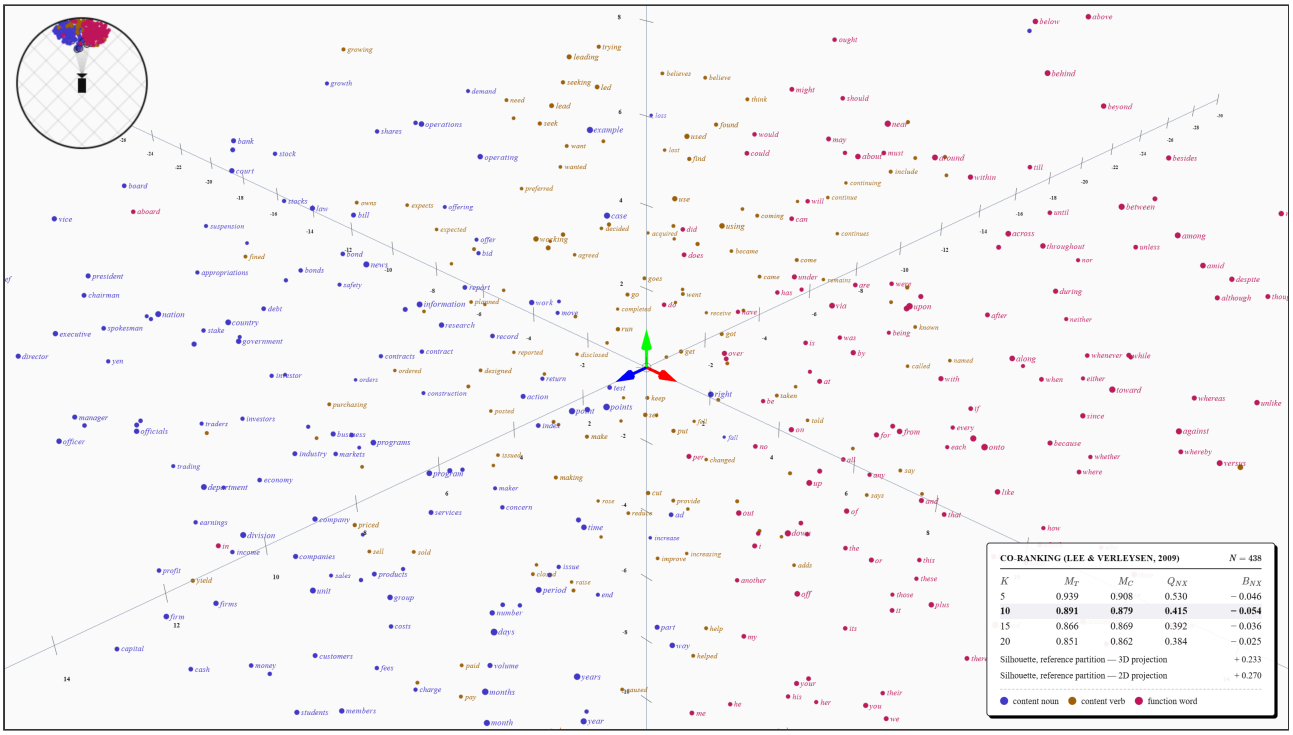


Fig. 9. Gemma 2 2B — t-SNE, perplexity 30.

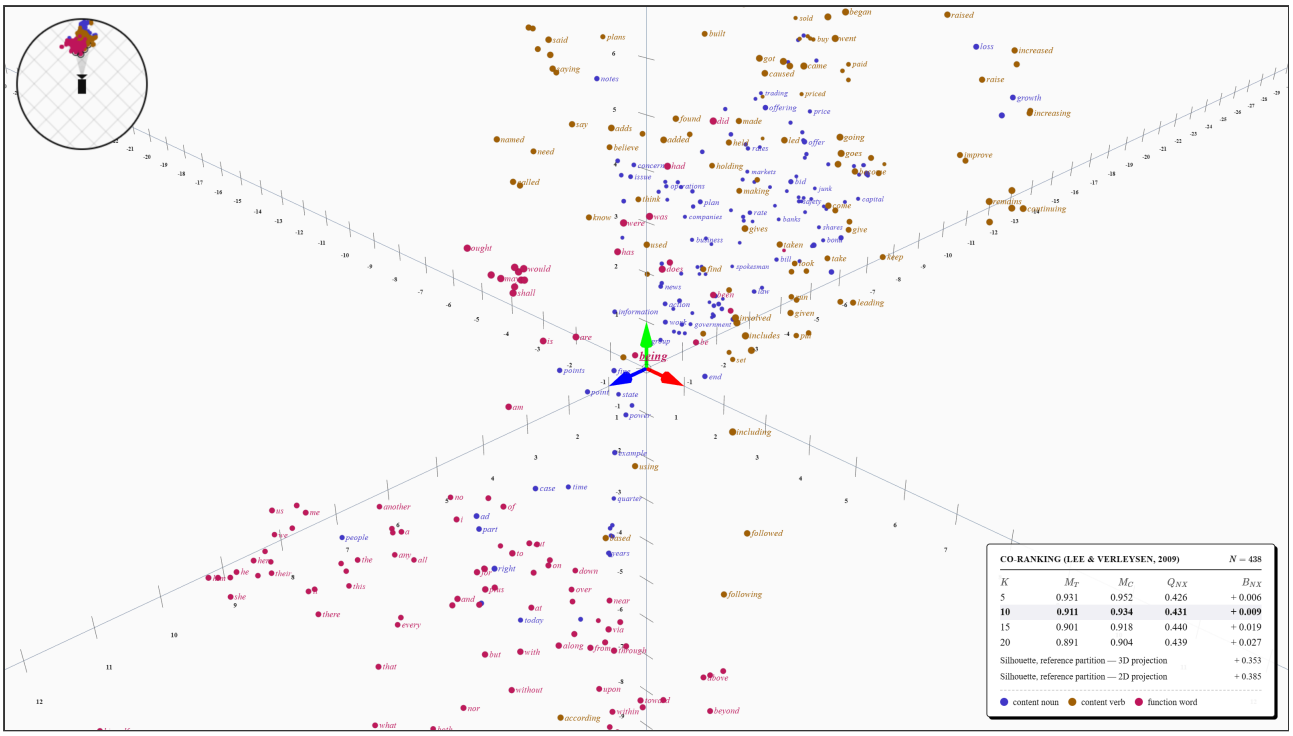


Fig. 10. Gemma 2 2B — UMAP.

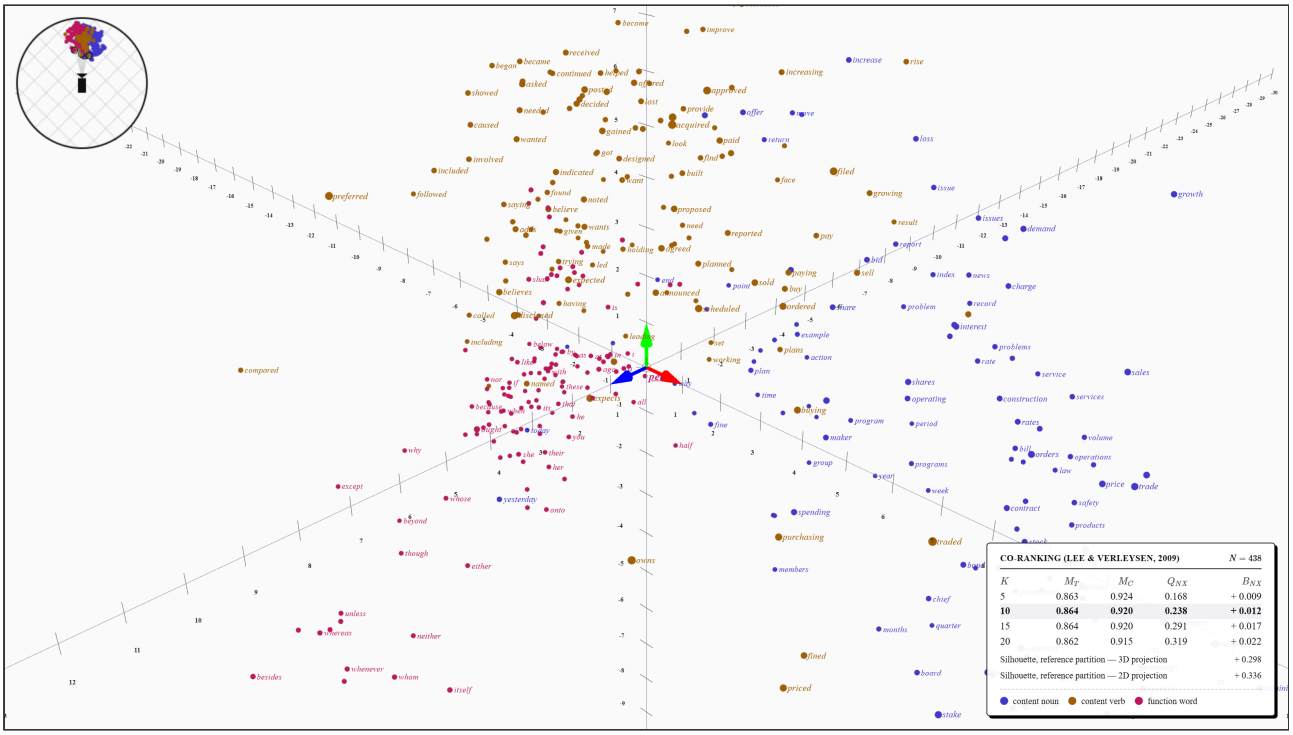


Fig. 11. Gemma 2 2B — Isomap.

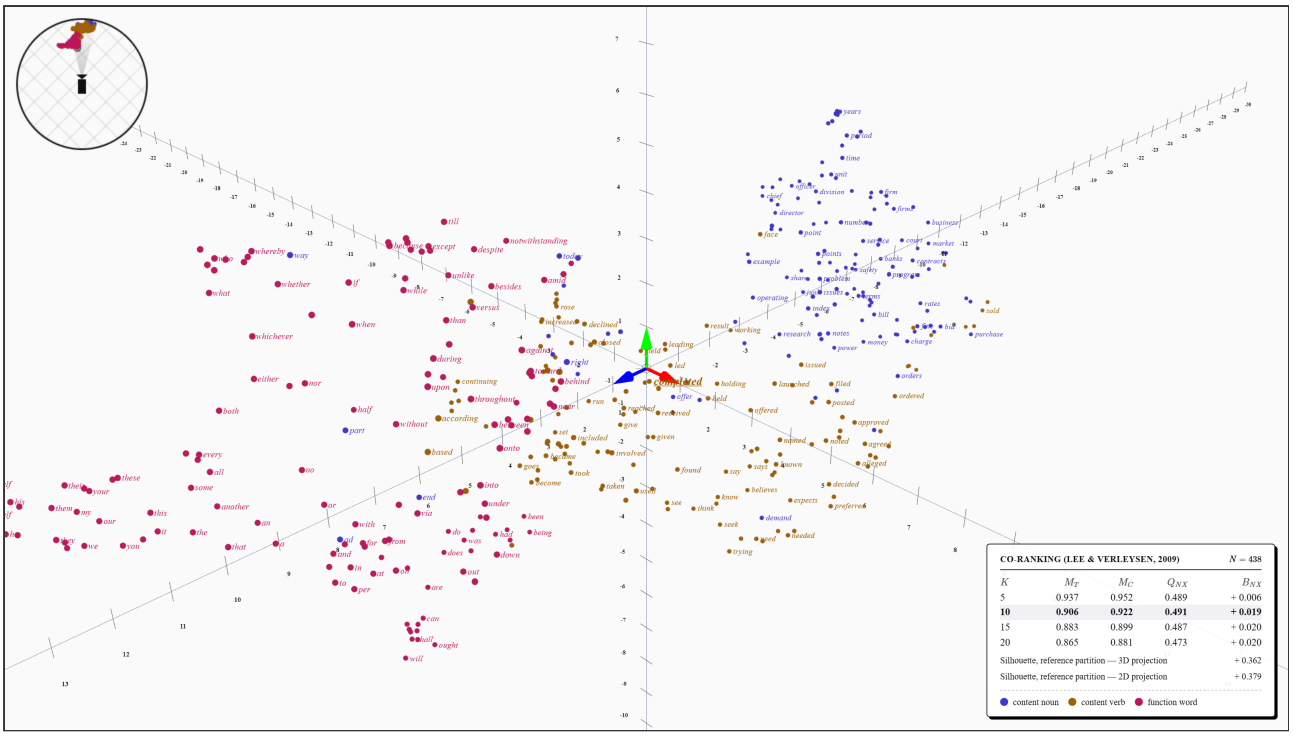


Fig. 12. Qwen 2.5 0.5B-Instruct — UMAP.

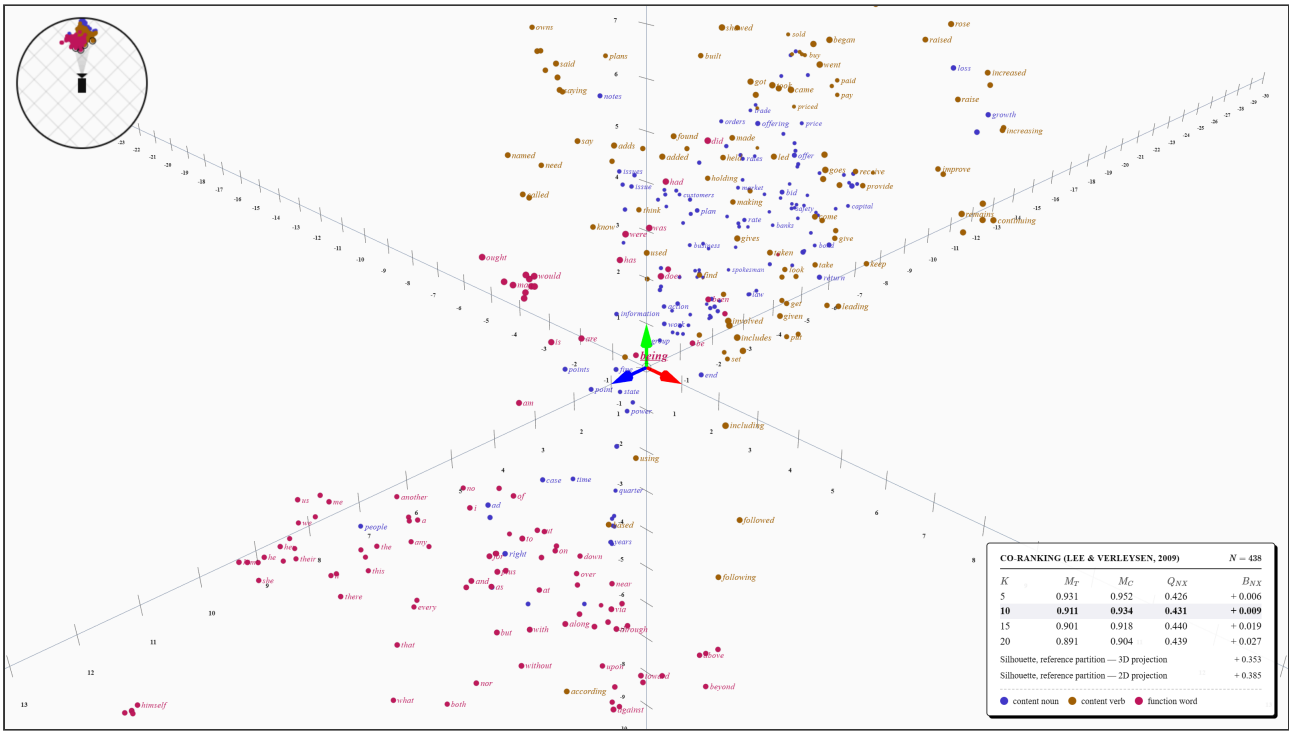


Fig. 13. Gemma 2 2B — UMAP.

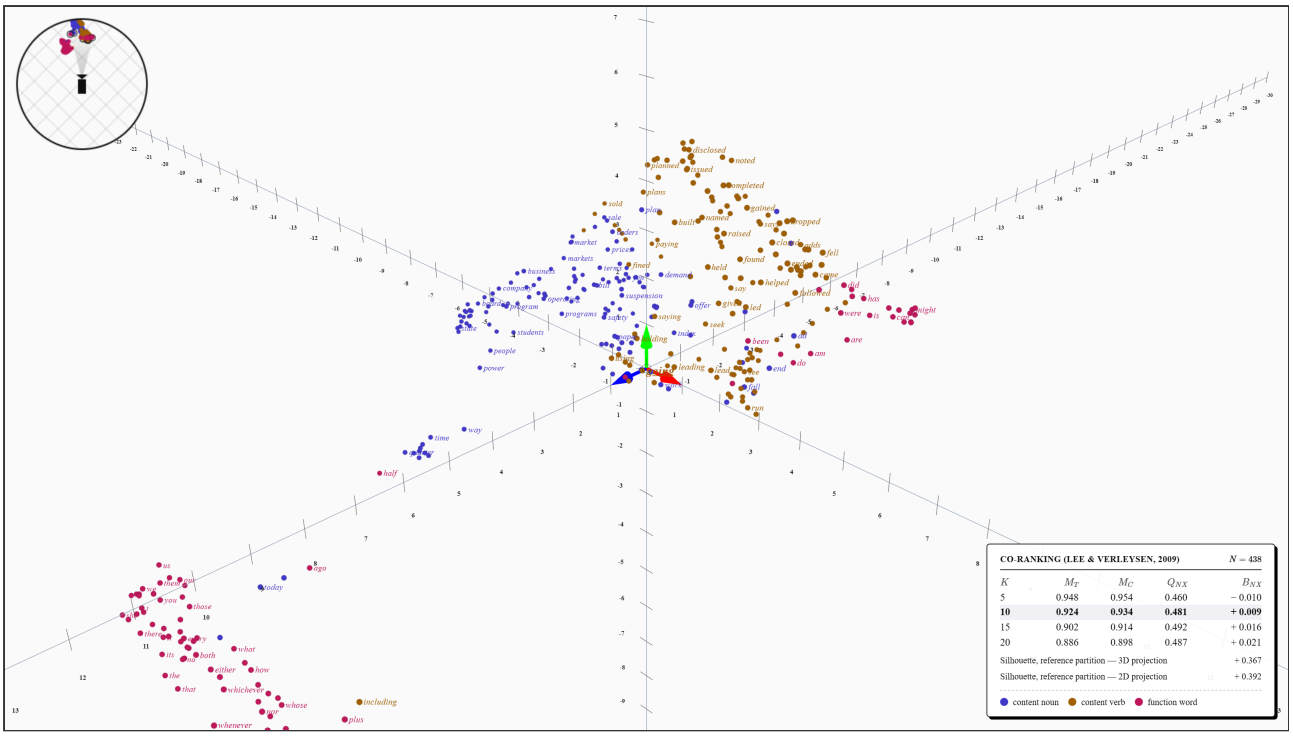


Fig. 14. GPT-2 — UMAP.

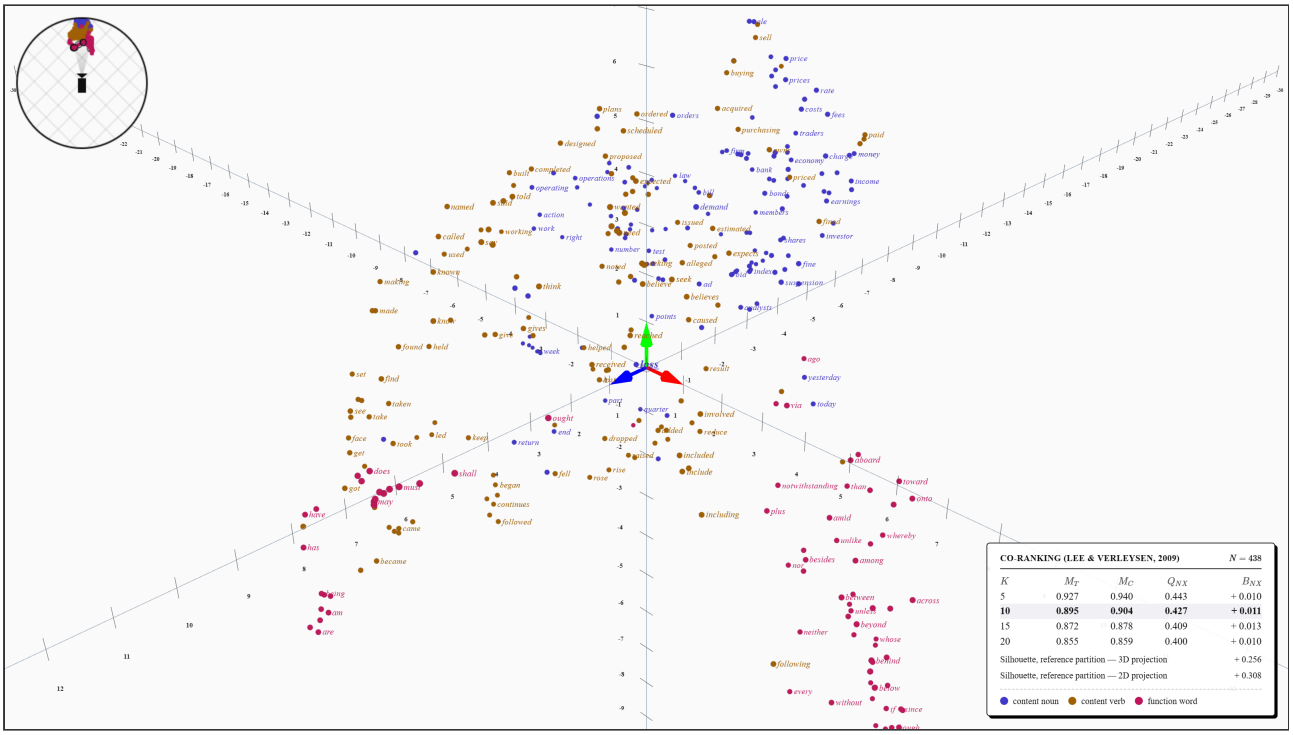


Fig. 15. BERT base uncased — UMAP.

5.4 Sensitivity to seed and perplexity

Model	Configuration	$M_T(10)$	$B_{NX}(10)$	s_{3D}	s_{2D}
Qwen 2.5 0.5B	t-SNE $p = 5$	0.878 ± 0.002	+0.012 ± 0.003	+0.249 ± 0.004	+0.297 ± 0.004
Qwen 2.5 0.5B	t-SNE $p = 30$	0.889 ± 0.005	−0.031 ± 0.007	+0.185 ± 0.028	+0.315 ± 0.025
Qwen 2.5 0.5B	UMAP	0.904 ± 0.004	+0.028 ± 0.005	+0.366 ± 0.008	+0.378 ± 0.006
GPT-2	t-SNE $p = 5$	0.896 ± 0.002	+0.022 ± 0.005	+0.243 ± 0.009	+0.312 ± 0.006
GPT-2	t-SNE $p = 30$	0.910 ± 0.004	−0.039 ± 0.009	+0.207 ± 0.022	+0.301 ± 0.022
GPT-2	UMAP	0.917 ± 0.003	+0.006 ± 0.009	+0.365 ± 0.009	+0.385 ± 0.018
Gemma 2 2B	t-SNE $p = 5$	0.883 ± 0.004	+0.018 ± 0.006	+0.228 ± 0.019	+0.289 ± 0.006
Gemma 2 2B	t-SNE $p = 30$	0.897 ± 0.003	−0.037 ± 0.006	+0.221 ± 0.015	+0.315 ± 0.027
Gemma 2 2B	UMAP	0.914 ± 0.002	+0.004 ± 0.011	+0.356 ± 0.007	+0.365 ± 0.011
BERT base	t-SNE $p = 5$	0.879 ± 0.001	−0.002 ± 0.003	+0.203 ± 0.009	+0.261 ± 0.008
BERT base	t-SNE $p = 30$	0.874 ± 0.004	−0.039 ± 0.005	+0.181 ± 0.019	+0.205 ± 0.036
BERT base	UMAP	0.891 ± 0.004	+0.017 ± 0.006	+0.255 ± 0.011	+0.295 ± 0.021

Mean ± sample standard deviation over seeds {0, 1, 2, 3, 42}. Source: `results/sensitivity_c2416895b599.json` *— commit* `e7f5a91f` *(clean working tree), Python 3.12.14, scikit-learn 1.6.1, umap-learn 0.5.7, seed 42, extraction: HTTP Range rows.*

5.5 Ablation: PCA without centring

Model	PCA M_T / M_C	PCA-NC M_T / M_C	PCA-NC B_{NX}
Qwen 2.5 0.5B	0.830 / 0.828	0.793 / 0.794	+0.000
GPT-2	0.827 / 0.846	0.761 / 0.799	−0.003
Gemma 2 2B	0.867 / 0.802	0.843 / 0.762	−0.008
BERT base	0.819 / 0.810	0.778 / 0.746	−0.009

6. Discussion

The sign of t-SNE's behaviour follows the perplexity. With the frozen perplexity of 5, t-SNE is mildly intrusive in three models and neutral in BERT; with perplexity 30 it is extrusive in all four, and the gap is many times larger than the spread across five seeds (Section 5.4). A statement such as "t-SNE tears the embedding space" is therefore a statement about a hyperparameter as much as about the algorithm. Both values lie in the range recommended by the method's authors, so neither can be dismissed; the defensible report is the pair. Low perplexity also showed a numerical artefact: with perplexity 5 a single word was occasionally ejected far from the cloud, a different one depending on the seed, although the word was not isolated in the original space.

Projections inflate category separation. In the original spaces the reference categories are barely separated (Silhouette 0.025–0.048), while every projection raises the score above its original-space value, in most configurations to 0.15–0.40. The dimension of the display matters too: for the non-linear methods the 2D Silhouette exceeds the 3D one in all but one

or two of the 16 configurations, whereas for PCA it is lower in three of the four models. A visually clean separation of nouns, verbs and function words in a projection is thus weak evidence about the embedding space itself; the co-ranking metrics are the appropriate check, and Silhouette values from 2D and 3D displays should not be compared directly.

Neighbourhood preservation depends on the algorithm. UMAP reaches the highest trustworthiness in every model. Isomap has the lowest or near-lowest trustworthiness — below PCA, whose three components capture only 8–11% of the variance, in three of the four models — but continuity well above it in every model, which B_{TC} reports as clearly intrusive ($+0.057$ to $+0.117$).

Faithfulness differs between models, independently of dimension. Under both t-SNE settings and UMAP, GPT-2 — the oldest and smallest of the four models, trained on a different corpus — has the highest trustworthiness, and BERT the lowest, although both embed tokens in 768 dimensions; under PCA, Gemma 2 is highest. Whatever makes an input-embedding space easier to project faithfully, it is not its dimension alone. As stated in Section 3.1, the design does not isolate which property of training is responsible.

Two behaviour indicators, two views. B_{NX} counts mild events only; B_{TC} weights hard events by their rank error. They can disagree: with perplexity 30, B_{NX} is negative for t-SNE in all four models, while the sign of B_{TC} differs between models. Reporting both avoids over-reading either.

6.1 Limitations

Static embeddings only. Contextual hidden states are deliberately out of scope (Section 2.1). **No causal attribution.** The models differ in many factors at once. **Sample.** The dataset comes from the NLTK sample of the Penn Treebank and is limited by the 146 function words that are single tokens in all tokenizers; the results describe these 438 frequent English word types. **Descriptive statistics.** The main matrix uses one seed per configuration; variability is reported by the sensitivity analysis, and no hypothesis tests are performed. **Reproducibility across platforms.** All numbers in Section 5 come from the continuous-integration run (Linux, clean checkout of the recorded commit); two such runs on different commits with unchanged pipeline code agreed to within 10^{-6}. A local re-run on Windows reproduced the dataset hash and the deterministic projections (PCA, PCA-NC, Isomap) exactly, but not t-SNE and UMAP: with the same seed, their metrics differed across the two platforms by amounts comparable to the seed-to-seed spread of Section 5.4. A fixed seed therefore guarantees reproducibility only on a fixed software and hardware stack, and the seed distribution, not a single seed, is the appropriate report for stochastic projections. **Single-tokenisation bias.** Restricting to single-token words favours frequent, short words, whose embeddings are typically better trained.

7. Conclusion

ENSy turns the projection of an embedding matrix from a picture into a measurement. Its co-ranking implementation is validated against an independent library, synthetic permutations and the primary source, and its results are regenerated from a frozen dataset by a pinned pipeline. Applied to four language models, it shows that the visual impressions most often drawn from such projections — whether a method crowds or tears the space, and whether linguistic categories form clusters — depend strongly on the projection and its hyperparameters. Projections of embedding spaces should therefore be reported with their neighbourhood-preservation metrics and their hyperparameters, never alone. Natural extensions are contextual representations, larger and multilingual samples, and inferential statistics over seeds and samples.

Appendix A: Reproducing the results

```
pip install -r requirements.txt          # pinned versions
export HF_TOKEN=...                       # licence accepted for google/gemma-2-2b
python -m ensy.dataset                    # rebuild data/dataset.json (hash must match)
python -m ensy.experiment                 # main matrix    -> results/main_<hash>.json
python -m ensy.experiment --ablation      # PCA-NC         -> results/ablation_<hash>.json
python -m ensy.experiment --sensitivity   # seeds/perplexity -> results/sensitivity_<hash>.json
python -m ensy.compare results/main_<hash>.json other/main_<hash>.json --markdown
python docs/update_results.py             # regenerate the tables and plots of Section 5
python docs/capture_screenshots.py        # regenerate the figures (server running)
python docs/md_to_pdf.py                  # build this PDF
```

Appendix B: Additional views

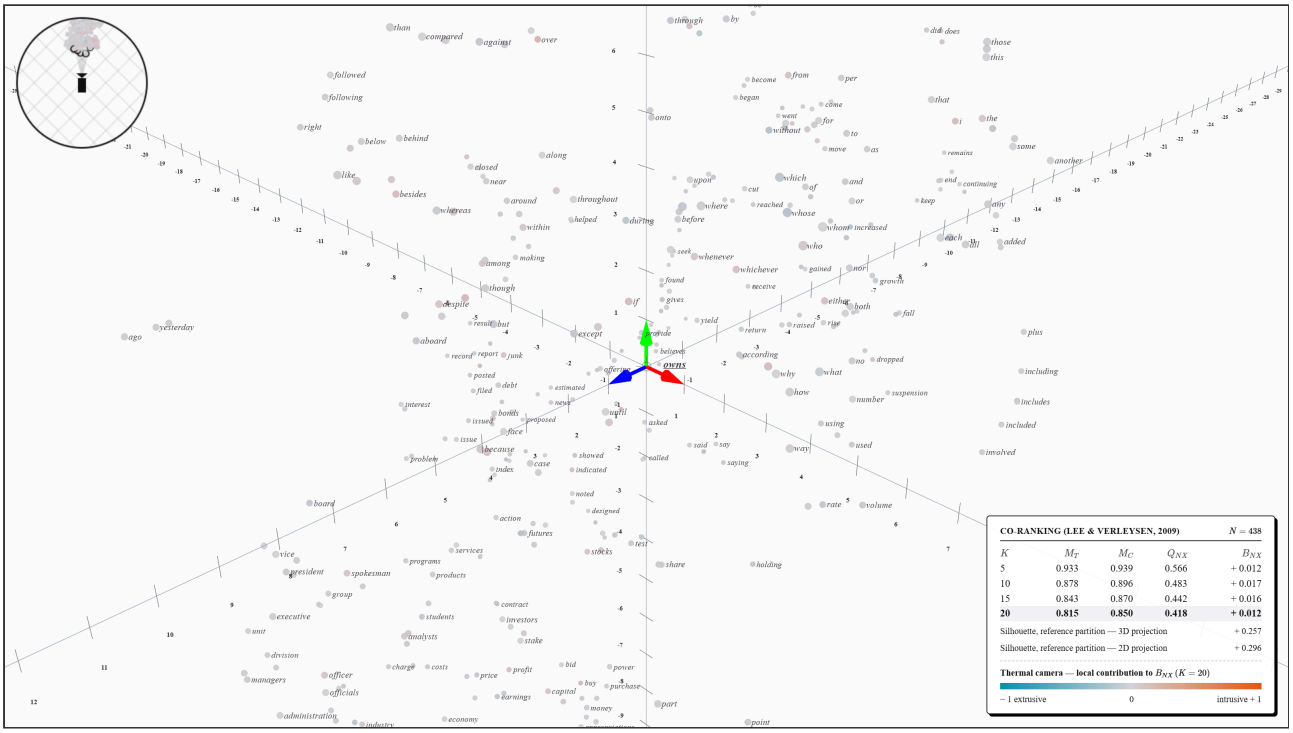


Fig. 16. Thermal camera for the same projection as Figure 2 at $K = 20$.

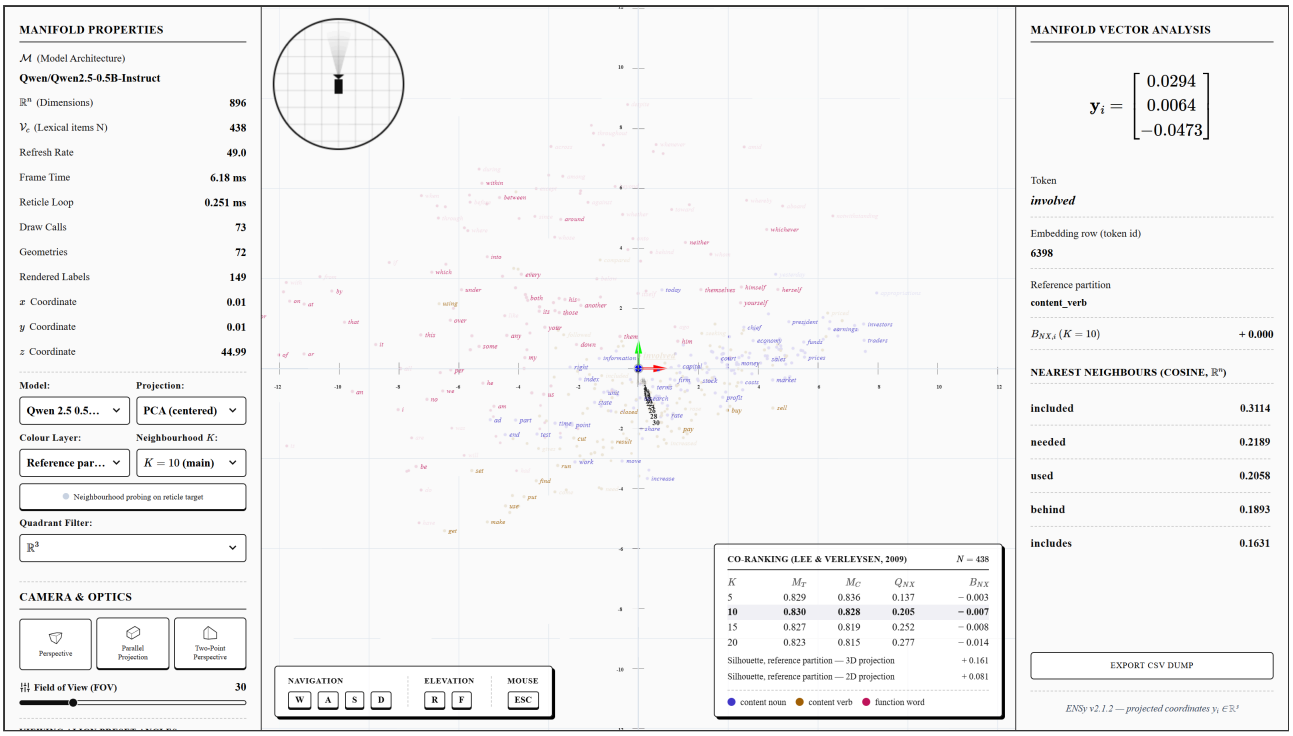


Fig. 17. Front (XY) camera preset on the PCA projection of Qwen 2.5.

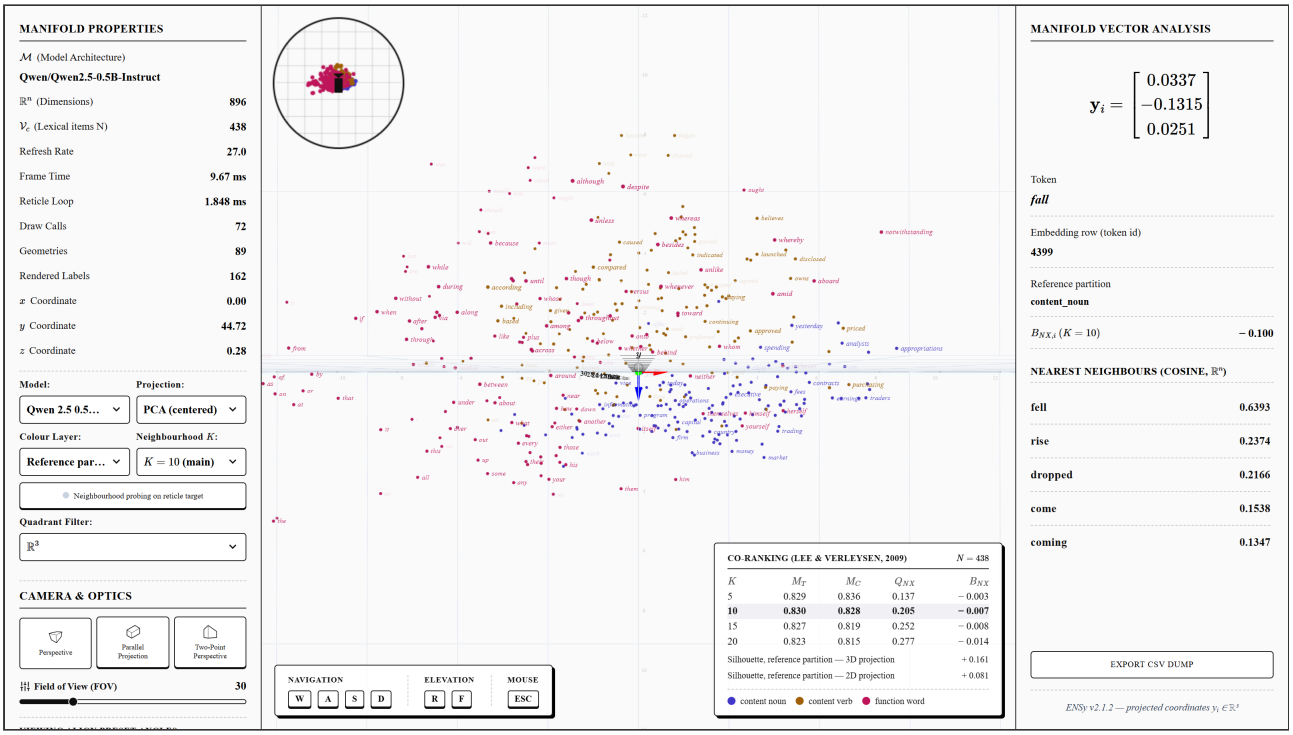


Fig. 18. Top (XZ) camera preset on the PCA projection of Qwen 2.5.

Appendix C: API

Endpoint	Description
`GET /api/health`	dataset path, N, frozen flag
`GET /api/models`	models, gating, whether the dataset covers them
`GET /api/dataset`	dataset metadata and items
`POST /api/dataset`	rebuild the dataset (refuses to overwrite a frozen one without `force`)
`GET /api/projection/{model}/{algorithm}`	coordinates, metric curves, $B_{NX,i}(K)$, neighbour lists, Silhouette (3D, 2D)
`GET /api/compare`	metric curves of every configuration of the main matrix

`model` ∈ {`qwen`, `gemma2`, `gpt2`, `bert`}; `algorithm` ∈ {`pca`, `pca_nc`, `tsne`, `tsne_p30`, `umap`, `isomap`}.

Appendix D: Keyboard controls

Input	Action
Click viewport	engage pointer lock (flight mode)
W / S	fly forward / backward along the aim direction (towards the reticle)
A / D	strafe left / right
R / F	move up / down
Mouse	look (pitch clamped)
Wheel	zoom
H	toggle the navigation HUD
Esc	release pointer lock

References

[1] J. A. Lee and M. Verleysen, "Quality assessment of dimensionality reduction: Rank-based criteria," *Neurocomputing*, vol. 72, no. 7–9, pp. 1431–1443, 2009.

[2] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," in *Proc. Int. Conf. Learn. Represent. (ICLR) Workshop*, 2013, arXiv:1301.3781.

[3] L. G. Nonato and M. Aupetit, "Multidimensional projection for visual analytics: Linking techniques with distortions, tasks, and layout enrichment," *IEEE Trans. Vis. Comput. Graphics*, vol. 25, no. 8, pp. 2650–2673, 2019.

[4] K. Ethayarajh, "How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings," in *Proc. EMNLP-IJCNLP*, Hong Kong, China, 2019, pp. 55–65.

[5] X. Cai, J. Huang, Y. Bian, and K. Church, "Isotropy in the contextual embedding space: Clusters and manifolds," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[6] K. Pearson, "On lines and planes of closest fit to systems of points in space," *Philos. Mag.*, vol. 2, no. 11, pp. 559–572, 1901.

[7] H. Hotelling, "Analysis of a complex of statistical variables into principal components," *J. Educ. Psychol.*, vol. 24, pp. 417–441 and 498–520, 1933.

[8] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.

[9] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," arXiv:1802.03426, 2018.

[10] J. B. Tenenbaum, V. de Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, 2000.

[11] J. Venna and S. Kaski, "Neighborhood preservation in nonlinear projection methods: An experimental study," in *Proc. Int. Conf. Artif. Neural Netw. (ICANN)*, Lecture Notes in Computer Science, vol. 2130, 2001, pp. 485–491.

[12] L. Chen and A. Buja, "Local multidimensional scaling for nonlinear dimension reduction, graph drawing, and proximity analysis," *J. Amer. Statist. Assoc.*, vol. 104, no. 485, pp. 209–219, 2009.

[13] J. A. Lee and M. Verleysen, "Quality assessment of nonlinear dimensionality reduction based on K-ary neighborhoods," in *Proc. Workshop on New Challenges for Feature Selection in Data Mining and Knowledge Discovery (FSDM)*, JMLR Workshop and Conf. Proc., vol. 4, 2008, pp. 21–35.

[14] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.

[15] Qwen Team, "Qwen2.5 technical report," arXiv:2412.15115, 2024.

[16] Gemma Team, "Gemma 2: Improving open language models at a practical size," arXiv:2408.00118, 2024.

[17] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," OpenAI, Tech. Rep., 2019.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[19] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38–45.

[20] M. P. Marcus, B. Santorini, and M. A. Marcinkiewicz, "Building a large annotated corpus of English: The Penn Treebank," *Comput. Linguist.*, vol. 19, no. 2, pp. 313–330, 1993.

[21] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. Sebastopol, CA, USA: O'Reilly Media, 2009.

[22] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[23] J. Venna and S. Kaski, "Local multidimensional scaling," *Neural Netw.*, vol. 19, no. 6–7, pp. 889–899, 2006.